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Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes --Manuscript Draft--

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Abstract:	Summary Horizontal gene transfer (HGT) challenges conventional Darwinian evolutionary models by introducing genetic material across species boundaries. Research on HGT in eukaryotes has focused on transfers of coding sequences (Villa & Viñas, 2019; Li et al., 2022), whereas insertions of non-coding DNA remain poorly understood. We investigated horizontally transferred DNA in the genomes of 14 cockroaches and four termites, including eight genomes newly sequenced using long-read technology. By focussing on transferred DNA inserts ≥ 50 bp, we found that cockroach genomes harbored between 101 and 4955 horizontal DNA transfers from the obligate endosymbiont <i>Blattabacterium cuenoti</i>. Previous studies have found little or no evidence for such horizontal transfers in eukaryotes harboring long-term endosymbionts. Some inserts appear to have persisted for >28.7 million years in panesthiine burrowing cockroaches. We identified numerous novel chimeric inserts comprising up to eight short segments from different locations in the <i>Blattabacterium</i> genome. RNAseq data from three species indicated that 5.04–8.58% of the inserts are transcribed in whole insect preparations, the majority of which (3.38–6.30% of all inserts) were from intergenic regions. The 4955 inserts found in <i>Macropanesthia lithgowae</i> exceed by >29 times the number of inserts (170) previously reported from

	any eukaryotic species, with the exception of rotifers. This level of DNA transfer is correlated to that of the transfers from the mitochondrial genome, an ancient bacterial endosymbiont turned organelle, highlighting the intimate relationship between endosymbiont and host genomes.
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23rd December, 2025

RE: Submission of manuscript for publication in *Current Biology*

Dear Editors,

We wish to submit an original research article entitled “**Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes**” for your consideration. The study is authored by Kyle Ewart, Maxim Adams, Zhuzhi Zhang, Louise Baker, Kokuto Fujiwara, Yoshinobu Hayashi, Leo Featherstone, Oscar Lo Lu, Jil Helbling, Maya Moral, Kiyoto Maekawa, Harley Rose, Aaron Jex, Simon Ho and Nathan Lo. We believe this manuscript would be suitable for publication as a ‘*Report*.’ This work has not been published elsewhere, nor is it currently under consideration for publication in any other journal.

Horizontal gene transfer (HGT) challenges conventional Darwinian evolutionary models by facilitating the movement of genetic material across species boundaries. Although HGT has been widely studied in eukaryotes, particularly in the context of bacterial endosymbionts, previous studies have generally identified relatively few transfers into host genomes (<170; with the exception of rotifers). Importantly, most of this work has focused on protein-coding genes; the extent and evolutionary significance of non-coding DNA transfer poorly understood.

In this study, we investigated HGT from the obligate bacterial endosymbiont *Blattabacterium cuenoti* into the genomes of its cockroach hosts. This symbiont is found in specialized cells of cockroach fat bodies and is inherited maternally through the eggs, providing the potential for recurrent transfer events. We analysed 14 cockroach and 4 termite genomes, including 8 newly generated long-read assemblies, and focused on putative HGT inserts ≥ 50 bp in both coding and non-coding regions. Using this approach, we identified tens of thousands of *Blattabacterium*-derived inserts across cockroach genomes (93–4900 per genome), orders of magnitude more than previously reported for any eukaryote outside rotifers. These inserts are typically short, and in some cases have persisted for more than 28 million years. We also identified numerous novel chimeric inserts composed of multiple segments originating from different regions of the *Blattabacterium* genome.

Together, our findings substantially revise current understanding of host–endosymbiont genome interactions by demonstrating pervasive horizontal transfer from a long-term obligate symbiont. The scale of transfer observed parallels that of mitochondrial DNA insertions (numts), highlighting similarities between contemporary endosymbiont–host interactions and the ancient bacterial endosymbiosis that gave rise to mitochondria. More broadly, this study provides a framework for uncovering previously hidden endosymbiont-derived DNA and advances our understanding of how horizontal transfer shapes eukaryotic genome evolution.

We would be pleased if you would consider our article for publication as a ‘*Report*’ in *Current Biology*. We look forward to hearing from you.

Kind regards,

Kyle Ewart (on behalf of all co-authors)
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Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes

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Summary

Horizontal gene transfer (HGT) challenges conventional Darwinian evolutionary models by introducing genetic material across species boundaries. Research on HGT in eukaryotes has focused on transfers of coding sequences (Villa & Viñas, 2019; Li et al., 2022), whereas insertions of non-coding DNA remain poorly understood. We investigated horizontally transferred DNA in the genomes of 14 cockroaches and four termites, including eight genomes newly sequenced using long-read technology. By focussing on transferred DNA inserts ≥ 50 bp, we found that cockroach genomes harbored between 101 and 4955 horizontal DNA transfers from the obligate endosymbiont *Blattabacterium cuenoti*. Previous studies have found little or no evidence for such horizontal transfers in eukaryotes harboring long-term endosymbionts. Some inserts appear to have persisted for >28.7 million years in panesthiine burrowing cockroaches. We identified numerous novel chimeric inserts comprising up to eight short segments from different locations in the *Blattabacterium* genome. RNAseq data from three species indicated that 5.04–8.58% of the inserts are transcribed in whole insect preparations, the majority of which (3.38–6.30% of all inserts) were from intergenic regions. The 4955 inserts found in *Macropanesthia lithgowae* exceed by >29 times the number of inserts (170) previously reported from any eukaryotic species, with the exception of rotifers. This level of DNA transfer is correlated to that of the transfers from the mitochondrial genome, an ancient bacterial endosymbiont turned organelle, highlighting the intimate relationship between endosymbiont and host genomes.

Results

Cockroach genomes contain thousands of Blattabacterium-derived inserts

We explored the prevalence, distribution and characteristics of DNA inserts in the genomes of cockroaches and termites that originated from the symbiont *Blattabacterium cuenoti* (hereafter *Blattabacterium*). This symbiont is found in specialized cells of cockroach fat bodies, and is inherited maternally through the eggs (Sacchi et al., 1988). *Blattabacterium* infected the ancestors of cockroaches >150–200 Mya (Lo et al. 2003; Bourguignon et al. 2018) and is present in almost all lineages, but was lost from all termites with the exception of *Mastotermes darwiniensis*. Although long-term obligate endosymbionts such as *Blattabacterium* are thought to contribute no or few DNA transfers from their genomes to host genomes, previous studies have only focused on the transfer of protein-coding sequences (Nikoh et al. 2010; Husnik et al. 2013; Luan et al. 2015; Sloan et al. 2014; Liu et al 2025).

To locate all possible DNA transfers from *Blattabacterium* to host genomes, we first created a library comprising 77 *Blattabacterium* genomes (Table S1), each of which was broken into 150-bp fragments. We then aligned these short *Blattabacterium* fragments to 23 genomes representing 14 cockroach and four termite species (multiple genome versions were used for some species) to characterize putative inserts with a minimum length of 50 bp. To identify unique insert events within a genome, we excluded inserts likely arising from duplication or transposition (based on the sequence similarity of inserts and their flanking regions).

Using this novel approach, we characterized 40,485 unique *Blattabacterium*-derived inserts across the cockroach genomes (93–4900 per genome; Fig. 1A), ranging in size from 50 to 5743 bp, with a median size of 161 bp (Fig. 1B). Most of the genomes in which we detected large numbers (i.e. several thousand) of *Blattabacterium* inserts were sequenced using ONT long-read technology, with N50s of 860–5330 kb, and 95.2–99.5% complete BUSCOs (Table S2). All inserts were found within long reads, flanked on both sides by cockroach host sequence (Fig. S1). Further, the long-read sequencing depth of these insert regions was 11.4–36.9x, indicating that the *Blattabacterium* and cockroach flanking DNA was not a result of artifactual chimeric reads.

We carried out additional analyses to confirm that the identified inserts were ‘true’ transfers from *Blattabacterium* to host, rather than assembly artifacts, or derived directly from *Blattabacterium* DNA. These included the removal of inserts arising from false assembly duplications, highly repetitive or low-complexity regions, and long regions aligning with bacterial DNA that may have been derived from genuine *Blattabacterium* sequence reads. Finally, manual inspection of BLAST results from inserts of each genome confirmed that they: 1) had been transferred from *Blattabacterium* and not other related Bacteroidales; and 2) were not the result of assembly errors.

Evolution of *Blattabacterium*-derived inserts in cockroach genomes

Inserts in host genomic DNA were found to originate from across the entire *Blattabacterium* genome, which is highly conserved in size, gene content, and gene order across cockroach species. There was no apparent bias in the origin of inserts with respect to their position in the *Blattabacterium* genome (Fig. 1C). In the five cockroach species with the highest number of inserts (>4000), the total amount of inserted DNA ranged from 687 to 1014 kb, representing 0.05–0.06% of total cockroach genome length. An examination of insert position across all species revealed that 15.38–25.56% were found in intragenic regions, and 74.44–84.62% were found in intergenic regions (excluding termites, in which we found only one insert). This differs somewhat from the proportion of genic regions in our assembled cockroach genomes, which was >25% in all but four genomes (range 13.79–33.71%). The average size of inserts (+/- standard deviation) in intra- and intergenic regions was 199.13 (+/- 138.75) bp and 191.70 (+/- 126.24) bp, respectively, which was significantly higher for the genic inserts (one-way nested analysis of variance: $F_{1,39424} = 17.96$, $p < 0.01$; Fig. 1B). The insert sequences comprised a comparable GC content (17.9–29.6%) to the *Blattabacterium* genomes (21.3–28.2%), which is considerably lower than the GC content of the cockroach and termite genomes (34.5–41.1%).

Closely related cockroach taxa were found to have similar numbers of inserts, suggesting that many of the inserts had been inherited in a vertical fashion (Fig. 1A). We inferred a minimum ‘age of transfer’ for each insert by calculating its similarity to inserts in other species. If an insert shared higher sequence identity with inserts in other cockroach species than to the *Blattabacterium* genome harbored by its host, we inferred that the insert was transferred in an ancestor of these species (Fig. S2). In contrast, inserts that displayed the highest identity with *Blattabacterium* genomes of their hosts were inferred to have been transferred after diverging from the other species included in the analysis.

We confirmed the homology of 22 putatively ancestral inserts, and thereby their common origin, by inspecting 5 kb of cockroach genome sequence upstream and downstream from the insert position across multiple taxa (Fig. 2A). For all examined inserts, although multiple taxa harboured homologous DNA across these ~10 kb regions, some species were inferred to have subsequently lost the inserts. Using this approach, and considering only inserts in homologous regions, we inferred that the oldest inserts were transferred into the ancestor of all panesthiine and geoscapheine species, which lived at least ~28.7 Mya (Beasley-Hall et al., 2021). There was also one potentially ancient *Blattabacterium* insert in the *Z. nevadensis* genome (Fig. 1A); however, we were unable to confidently verify its authenticity with our methods, and it thus requires further study. We also examined pairwise nucleotide divergence within inserts and their flanking regions (500 bp upstream and downstream, or shorter when inserts were near scaffold ends) for 22 putatively ancestral inserts. Divergence within inserts was not consistently lower than that of adjacent regions, providing no clear evidence for insert function.

Transcription and chimeric nature of inserts

We used RNAseq data to investigate the transcription of inserts in *Panesthia t. tegminifera* (2011), *Macropanesthia lithgowae* (2009), and *Panesthia sloanei* (2001). By mapping insert

131 sequences to RNA contigs using BLASTn, we found that 91.42–94.96% of inserts showed no
132 evidence of transcription. This likely represents a minimum estimate of the number of
133 transcribed inserts, as we did not survey RNA from all life stages and tissues. Of the remaining
134 inserts, which were observed in assembled RNA contigs, 26.59–32.97% overlapped with
135 annotated genic regions of the cockroach genome (1.66–2.28% of total inserts), while the
136 remainder overlapped with intergenic regions (3.38–6.30% of total inserts). A small fraction
137 (0.40–0.65%) of inserts mapped to exon sequences within RNA sequences.

138
139 During manual BLAST inspections, we noted that several inserts displayed a chimeric nature,
140 comprising *Blattabacterium*-like sequences of length 30–2940 bp from distant areas within
141 the *Blattabacterium* genome placed directly adjacent to each other (Fig. 2B). Such chimeric
142 DNA inserts were present in all cockroach genomes, independently sequenced by separate
143 research groups using different sequencing technologies. A maximum of nine
144 *Blattabacterium*-derived segments from distant genomic regions were found directly adjacent
145 to each other. We mapped four chimeric inserts against transcriptome data for *M. lithgowae*
146 and *P. tryoni tegminifera* and found that none of these were clearly transcribed, although two
147 had partial matches with RNA contigs.

148 149 150 Discussion

151
152 Our analyses have revealed large numbers of endosymbiont DNA transfer events within single
153 cockroach genomes. The number of HGTs that eventually become fixed in a species is
154 expected to represent only a small fraction of the actual number of acquisition events that
155 occur across evolutionary time (Van Etten & Bhattacharya, 2020). While there is an abundant
156 body of research focused on insertions with homology to protein-coding sequence, transfers
157 that result in sequences that do not encode proteins, or that result in sequences of unknown
158 function, have been largely ignored. Such insertions nevertheless introduce abrupt genetic
159 variation, which can persist in the genome if these insertions have minimal or no fitness
160 effects, if purifying selection is insufficient, or if they acquire functional roles such as
161 regulation of gene expression (Husnik & McCutcheon, 2018).

162
163 With the exception of rotifers, which have an unusually high number of transfers associated
164 with anhydrobiosis (Simion et al 2021), the highest number of transfers reported in any animal
165 was previously 170, in the whitefly *Bemisia tabaci* (Li et al 2022). There are a number of
166 reasons why we may have identified >25 times this number in the cockroaches that we
167 examined. First, we analyzed both intra- and intergenic regions of the genome. Second, while
168 previous studies have filtered for the completeness of inserted coding sequences, we
169 identified inserts as short as 50 bp. Relatively short sequences were indeed the most prevalent
170 type of insertion (median length = 161 bp). Third, previous studies often relied on short-read
171 sequencing, where inserts spanning the majority or entirety of a sequence read may have
172 been misidentified as bacterial contamination and discarded during genome assembly. In
173 contrast, we used long-read sequencing for most Panesthiinae genomes, which enabled us to
174 confirm the presence of these inserts within single sequence reads. We found differing
175 numbers of inserts in the separately published genomes of each of the pests *Periplaneta*
176 *americana* (101–658) and *Blattella germanica* (278–285), possibly due to the use of short-
177 read versus long-read sequencing technology in each case.

Fourth, we focused on horizontally transferred DNA segments from *Blattabacterium*, an obligate and maternally transmitted endosymbiont whose genome is likely to come in close contact with host nuclear DNA during egg formation and zygote development. Previous studies have generally recorded very few or no such transfers from long-term obligate endosymbiotic bacteria to the genomes of their insect hosts (Nikoh et al. 2010; Husnik et al. 2013; Luan et al. 2015; Sloan et al. 2014). This has been attributed to the lack of mobile genetic DNA in these genomes, as well as a lack of proteins associated with HGT encoded in their genomes (Liu et al., 2025). A recent study of HGT in two cockroach and 45 termite species did not detect any horizontal transfers from *Blattabacterium* in either of the two cockroaches examined (*Cryptocercus meridianus* and *Blatta orientalis*), or the termite *Mastotermes darwiniensis*, all three of which harbour this symbiont (Liu et al., 2025). However, this lack of detection of transferred DNA of symbiont origin is likely due to a focus on protein-coding sequence. We expect that the use of the approaches described in our study will result in increased detection of endosymbiont-derived inserts in the genomes of eukaryote hosts.

Panesthiine taxa were found to possess substantially higher numbers of *Blattabacterium*-derived inserts (Fig. 1A). This group have relatively large genomes. They are also known to have long generation times relative to most other cockroaches (Bell et al. 2007), which might be expected to reduce opportunities for transfer per unit time, if transfer occurs only during oogenesis. However, it is unknown whether and how other components of *Blattabacterium* life history, including population load across cockroach species, and rates of cell division, influence the prevalence of transfers. Microscopic analyses of *Blattabacterium* in host tissues have revealed high rates of cell division, in addition to frequent asymmetric cell division, compared with other insect symbionts (Noda et al., 2024). Alternatively, we previously found that Panesthiinae have experienced considerably more genome-wide relaxed selection than other cockroaches, including those examined here (Ewart et al., 2024). Insufficient purifying selection could possibly allow larger numbers of inserts to persist in Panesthiinae genomes, under the assumption that most inserts have neutral or slightly deleterious impacts on fitness. The function of *Blattabacterium*-derived DNA inserts in cockroach genomes remains unclear, but the persistence of ancestral inserts in the Panesthiinae, inferred to have been transferred >28.7 Mya, suggests that at least some of them have functional roles. The low number of inserts identified in termite genomes may reflect the long-term absence of *Blattabacterium*, as well as a considerable reduction in genome size compared with cockroaches.

The widespread occurrence of horizontally transferred DNA in cockroach genomes resembles the well-established phenomenon of mitochondrial DNA transpositions into the nuclear genome (i.e., numts; Pamilo et al., 2007; Uvizl et al., 2024; Keeling, 2024). We repeated our analyses using 150-bp fragments of cockroach and termite mitochondrial genomes, and detected numt inserts in each genome, ranging in number from 27 to 1095 (termites 27–150; cockroaches 251–1095; Table S3). The number of *Blattabacterium*-derived inserts was significantly correlated with the number of mitochondrial-derived inserts (Spearman's correlation, $p = 2.6 \times 10^{-5}$). The occurrence of numts is understood to be the outcome of primarily neutral processes, which may further implicate relaxed selection as a key factor regulating the tempo of insertion and persistence of exogenous DNA in the genome.

Our discovery of chimeric insertions comprising sequences from diverse parts of the *Blattabacterium* genome (e.g., see Fig. 2B) is akin to previous reports of chimeric numts in *Arabidopsis thaliana* (Fields et al., 2022). These chimeras are likely produced by non-homologous end-joining repair that incorporates segments of *Blattabacterium* DNA that possibly arise following symbiont cell replication or death. Chimeric genes have previously been described in other species, although at a broader scale than the chimeric inserts that we have identified in cockroaches (Rogers et al., 2012). In *Drosophila*, chimeric genes resulted from retrotransposition and the combination of exons and introns from two ‘parental’ genes (Long and Langley, 1993; Rogers et al 2009). The chimeric genes believed to have arisen in the ancestor of eukaryotes (Méheurst et al., 2018) are hypothesised to have come about as a result of intron invasion and combination of domains that allowed novel functions.

Our findings suggest that horizontal transfer of DNA from prokaryote symbionts to eukaryotes may be considerably more pervasive than previously thought. Furthermore, the persistence of numerous inserts over millions of years indicates that they may have assumed functional roles in both genes and intergenic regions, are effectively neutral, or are only slightly deleterious. Future research on cockroaches and other species harboring obligate endosymbiotic prokaryotes will assist with uncovering the potential functional effects of inserts, providing a more comprehensive understanding of how horizontal transfer shapes genome evolution.

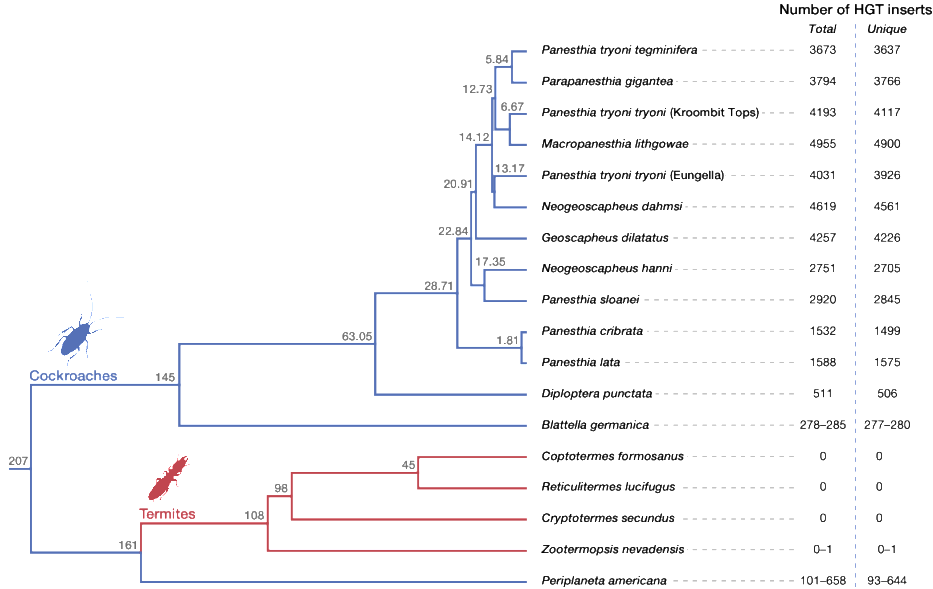
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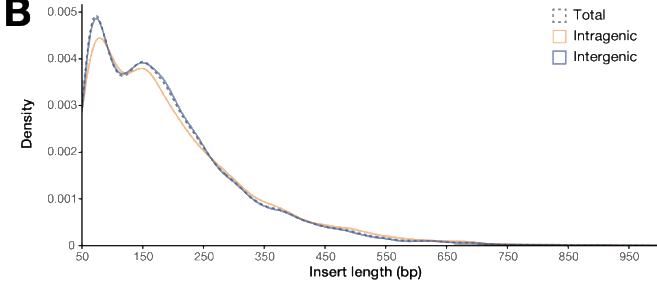
Declaration of interests

The authors declare no competing interests.

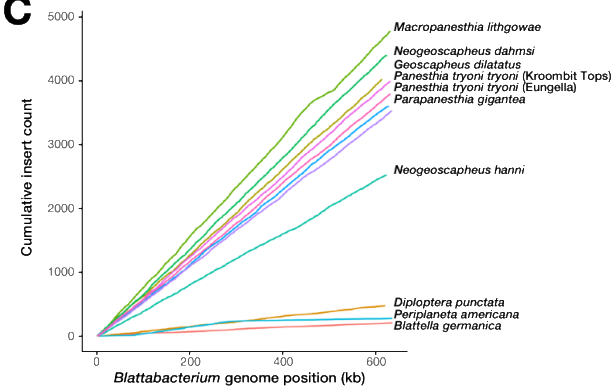
A



B



C



259 **Figure 1.** Chronogram of taxa analysed in this study alongside the total number of HGT inserts within each genome, and the number of unique
260 inserts (after excluding inserts likely arising from duplication or transposition) (A). A range of insert numbers are listed for species in which
261 multiple genomes were analysed. Node ages (Mya) are based on previous studies (Evangelista et al., 2019; Beasley-Hall et al., 2021; Adams et
262 al., 2024). The length distribution of inserts, further divided by genic and intergenic inserts (B). Only lengths >50 bp are depicted, as this was the
263 minimum length threshold, and the plot removed insert >1000 bp for clarity (51 of the 39,411 markers have been excluded). The origin of inserts
264 along the *Blattabacterium* genome based on the start position of the insert (C). The cumulative number of inserts is plotted along the
265 *Blattabacterium* genome with the linear trends indicating random origin of inserts from the *Blattabacterium* genome. Steeper curves correspond
266 to more inserts. Only cockroach species with their host host-specific *Blattabacterium* strain published were included in the analysis (see Table
267 S1). For species whose genomes have been sequenced multiple times by different research groups, only the genome version with the most inserts
268 was analysed (for both 'B' and 'C').
269

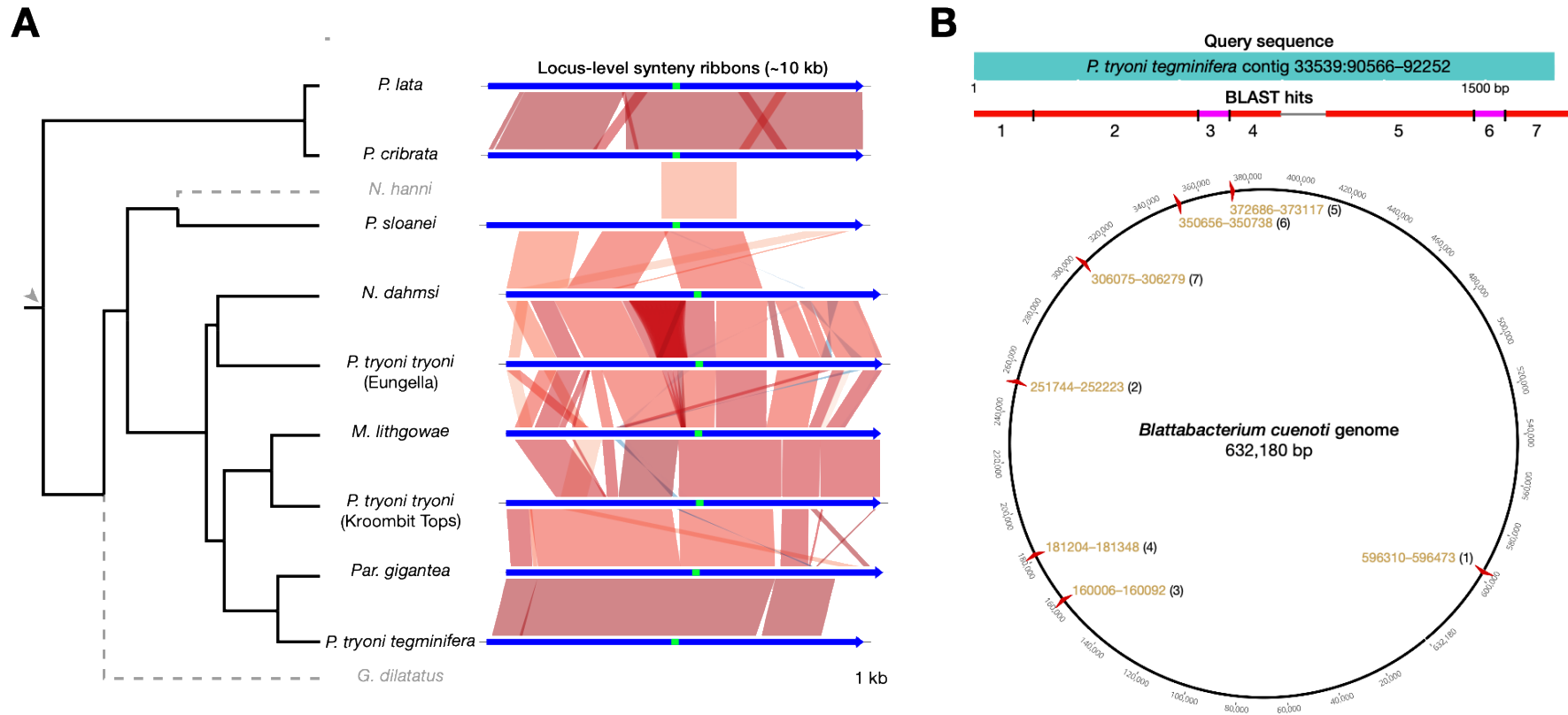


Figure 2. Example of how single putative ancestral insert was subsequently evaluated by examining the homology of its flanking regions (A). The right panel shows the homology of each insert (green bar) and its 5 kb flanking regions (blue bars) across the species in which the insert was identified, as indicated by the phylogenetic tree on the left (species lacking the insertion are shown with dashed lines). The red arrow indicates the inferred timing of this example HGT insertion. Example of a chimeric HGT insertion in the *Panesthia tryoni tegminifera* genome (B). The *P. t. tegminifera* contig, displayed at the top, is divided into seven segments of different origin, and the circular *Blattabacterium cunenoti* genome (CP142624.1), displayed at the bottom, indicates the source of these insertions. Numbers below the insertions indicate the corresponding positions in the cockroach and *Blattabacterium cunenoti* genomes (with yellow labels marking specific coordinates).

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STAR Methods

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

This study involves sequencing and assembling genomes from 8 Blaberidae species, analysed with 15 previously generated Blattodea and 77 *Blattabacterium* genomes (Table S1). Specimens of *Macropanesthia lithgowae*, *Neogeoscapheus dahmsi*, *Panesthia lata*, *Panesthia sloanei*, *Panesthia tryoni tegminifera*, *Panesthia tryoni tryoni* (Eungella), *Panesthia tryoni tryoni* (Kroombit Tops) and *Parapanesthia gigantea* were collected throughout Australia. Leg muscle and fat body tissue were used for DNA extractions (see Method Details).

METHOD DETAILS

DNA extraction, library and sequencing details

Genomic DNA was prepared from the leg muscle tissue of *M. lithgowae*, *P. tryoni tryoni* (Kroombit), *P. sloanei*, and *P. lata*, while the DNA of *Par. gigantea*, *P. tryoni tegminifera*, *P. tryoni tryoni* (Eungella) and *N. dahmsi* was extracted from both leg muscle and fat body tissue. Live cockroaches were chilled at -30 °C for 4–5 min to induce torpor, and the femoral muscle (~3 mm) was dissected from one or two hind legs, depending on body size. Following this, insects were frozen at -80 °C. Tissue was incubated overnight at 56 °C in 600 µl SDS buffer with 20 µL of 20 mg/mL stock Proteinase K (final concentration 400 µg/mL), followed by RNase A treatment (final concentration 20 µg/mL) at 37 °C for 1 hr. DNA extractions were performed using the phenol chloroform method, or using the NEB Monarch HMW DNA Extraction Kit (NEB #T3060). The phenol chloroform method involved two extractions with an equal volume of phenol:chloroform:isoamyl alcohol 25:24:1), isopropanol precipitation with sodium acetate (3 M, pH 5.5). The pellet was washed twice with 70% ethanol air-dried, and the DNA pellet resuspended in 40 µl of 10 mM Tris (pH 8.0). The NEB Monarch HMW DNA Extraction Kit was used according to the manufacturer's instructions, with the exception that Proteinase K digestion was undertaken for 4 hours at 600 rpm, rather than the recommended 45 minutes, to ensure thorough digestion of the tissue. As minimal gDNA appeared to stick to the glass beads, the supernatant was centrifuged for 10 minutes at 16,000 g at room temperature, the pellet was washed twice in 500 µL kit Wash Buffer, dried briefly then resuspended in 20 µL Elution Buffer II. The quality and integrity of the gDNA was assessed by a 4200 TapeStation System (Agilent Technologies) using Genomic DNA ScreenTape (Cat No. 5067-5365) and reagents (Cat No. 5067-5366). The absorbance ratios were assessed by Nanodrop Spectrophotometer (DeNovix DS-11). Concentration was determined using a Qubit 2.0 Fluorometer (Invitrogen by Life Technologies) using a dsDNA BR assay kit (Cat No. Q32850).

Long-read sequencing was performed using gDNA from *M. lithgowae*, *N. dahmsi*, *P. lata*, *P. sloanei*, *P. tryoni tryoni*, *P. tryoni tegminifera* and *Par. gigantea* (Table S1). Libraries were prepared from gDNA using the Ligation Sequencing Kit V14 (SQK-LSK114) protocol according to the manufacturer's instructions with the following modifications and options. The DNA Repair and End-Prep reaction continued for 15–20 minutes. The Adapter ligation reaction proceeded for 20 minutes and the kit LFB

wash buffer was used to enrich for longer fragments. Final library elution occurred at 37°C to improve the recovery of long fragments. The integrity and concentration of final libraries were assessed using a 4200 TapeStation System (Agilent Technologies) and Qubit 2.0 Fluorometer (Invitrogen by Life Technologies) as previously described.

Libraries were sequenced on a PromethION 24 sequencer (Oxford Nanopore Technologies) using R10.4.1 flow cells (FLO-PRO114M), following the manufacturer's instructions. Libraries were stored at 4°C for less than a week if not sequenced immediately. The sequencing output was additionally increased by washing the flow cells (EXP-WSH004) after approximately 24 and 48 hours, and reloading the libraries for sequencing for a total of 72 hours. Sequencing was performed in super-accurate base-calling v4.3.0 mode (400 bps).

Genome assembly and annotation

For the ONT sequencing that incorporated duplex base calling (Table S1), redundant simplex reads (i.e. simplex reads which have duplex offsprings) were removed using samtools v1.2 (Danecek et al., 2021). Adapters were trimmed for all ONT sequence data (with and without duplex calling) and converted into fastq files using dorado v0.6 (<https://github.com/nanoporetech/dorado>).

De novo assembly was carried out using the trimmed ONT data using Flye v 2.9.3 (Kolmogorov et al., 2019). After some initial optimisation experiments, the ONT sequencing data with duplex calling was assembled using the 'ONT corrected' mode (--nano-corr) and a minimum overlap of 10,000 (--min-overlap 10000), while the other ONT sequencing data was assembled using the 'ONT raw' mode (--nano-raw) and default settings. An additional two iterations of polishing were carried out after the assemblies. The purge_dups v1.2.5 (Guan et al., 2020) pipeline was used to remove putative haplotypic duplications and heterozygous overlaps from the genomes. Raw ONT reads were mapped to the draft assembly using minimap2 v2.18 (Li, 2018) for this purging step. QUAST v5.3.0 (Gurevich et al., 2013) was used to assess the assembly continuity, and BUSCO v5.8.1 was used to assess assembly completeness (utilizing the Insecta lineage data set; Simão et al., 2015; Waterhouse et al., 2019).

The genome assemblies were annotated using FGENESH++ v7.2.2 (Salamov & Solovyev, 2000; Solovyev et al., 2006), performed on the Nimbus cloud service provided by the Pawsey Supercomputing Centre. Prior to annotation, we generated a repeat library with RepeatModeler v2.0.1 (Flynn et al., 2020) and masked repetitive elements in the assemblies using RepeatMasker v4.0.6 (Smit et al., 2015), applying the '-nolow' option to exclude low-complexity or simple repeats from masking. The masked and unmasked assemblies were used as input for FGENESH++. Following Ewart et al. (2024), FGENESH++ was run with the non-mammalian general pipeline settings, employing optimized gene-finding parameters trained on the pea aphid (*Acyrtosiphon pisum*) genome, and a curated insect protein database (provided by Softberry; 2020 release) for "prot_map" homology-based predictions. *Ab initio* predictions lacking BLAST support were excluded.

These newly assembled genomes were combined with other cockroach and termite genomes from InsectBase 2.0 (Mei et al., 2022), NCBI and Ewart et al. (2024) for

subsequent analyses (see Table S1 for species/genome details). *Periplaneta americana*, *Blattella germanica*, and *Zootermopsis nevadensis* had multiple available genomes from various sources, which were all included to assess whether differences in data type and/or assembly methods affected the results in subsequent analyses.

QUANTIFICATION AND STATISTICAL ANALYSIS

Characterising HGT inserts

The following steps were carried out to identify, filter, and age *Blattabacterium cuenoti* (hereafter *Blattabacterium*) inserts in the cockroach and termite genomes. This pipeline workflow and the relevant scripts are available on GitHub (https://github.com/MEEP-projects/HGT_pipeline), and outlined in Fig. S3. Additionally, alternative thresholds for many of the steps were applied, and are included in brackets below (see the 'HGT insert QC' section). The corresponding results when applying these alternative thresholds are presented in the 'HGT insert QC' section.

We first created a library comprising 77 *Blattabacterium* genomes (see Table S1 for genome details), each of which were broken into 150 bp fragments (alternate threshold = 100 bp) using a custom AWK script. These *Blattabacterium* 'fragments' were aligned to the reference host genome using bwa mem v0.7.17 (Li & Durbin, 2009), implementing default parameters. The resultant alignment was converted into a bam file with samtools, then into a bed file using bedtools v2.3 (Quinlan and Hall, 2010). Regions of the cockroach genome against which *Blattabacterium* fragments aligned with were identified as potential horizontal inserts. In the case that aligned *Blattabacterium* fragments overlapped on a host contig, or aligned fragments were found within 150 bp of each other on a host contig (alternate threshold = 50 bp), we merged the host region aligning to these sequences as a single putative HGT insert. Putative inserts shorter than 50 bp (alternate threshold = 30 bp) were removed using AWK. Note that inserts were detected in genomes of species for which only leg muscle tissue was included during DNA extraction (thus excluding *Blattabacterium*), and also species for which fat body was included (thus including *Blattabacterium* cells and potentially genomic contigs).

Several filtering steps were carried out to remove potential artefactual HGT inserts. First, we conservatively removed any contigs in which genuine *Blattabacterium* sequences may have been combined with cockroach sequences during genome assembly (and therefore erroneously characterized as HGT inserts). Any contigs for which the total length of putative bacterial sequences (identified using the method above) was $\geq 20\%$ of the total contig length were removed. In addition, we conservatively removed any contigs with one or more contiguous putative HGT inserts above 10,000 bp in length (which were found to be significant outliers based on the length distributions of all putative inserts in our dataset; see Fig. 1B). Second, inserts that formed part of false assembly duplications were identified and removed. To identify these potential false duplicate inserts, the insert regions were extracted from the genomes with 300 bp of flanking sequence (to either side) using bedtools, then BLASTed against all of the other inserts (with flanks) within the same genome using blastn v2.2.3. Inserts were considered false duplicates and removed if the resultant

sequence identity was >90% (alternative threshold >95%), and query coverage was >90% (alternative threshold >95%). Third, highly repetitive low-complexity ("low-entropy") sequences were identified using the symmetrical DUST algorithm (Morgulis et al., 2006) implemented in Minimap2 v.2.28 (Li, 2018, 2021), and any inserts with contiguous, low-entropy regions representing $\geq 50\%$ of the total insert length were removed. Fourth, to ensure that we could characterise the cockroach genomic sequence surrounding each insert, we excluded inserts where either of the 300 bp flanks consisted of $\geq 50\%$ ambiguous (N) nucleotides. Lastly, inserts that were more similar to other bacteria species (i.e. not *Blattabacterium*) were removed, as they could be potential artefacts, or genuine inserts of other related Bacteroidales. Each insert was BLASTed against two custom databases: one comprising only *Blattabacterium* sequences database ($n = 77$; Table S1), and one with other bacteria, with *Blattabacterium* sequences removed ($n = 2,009,606$). If the sequence similarity of the bacteria database BLAST was >2% higher than the *Blattabacterium* database BLAST, then the sequence was removed. The threshold of 2% was chosen based on preliminary assessment of pairwise genetic divergence between *Blattabacterium* and close relative Bacteroidales.

Some of the inserts present in a host genome may be the result of a duplication or transposition of a previously inserted *Blattabacterium* sequenced. To identify the number of unique HGT inserts within a genome (i.e. to only count those arising from separate transfers from a *Blattabacterium* genome), we applied a similar approach to that used for the false duplication filter above: inserts were extracted with 300 bp of flanking sequence, then BLASTed against all of the other inserts (with flanking) within the same genome. HGT inserts with a sequence identity >70% (alternative threshold >55%), and query coverage >70% (alternative threshold >55%) were considered duplicated. Finally, manual auditing of the results was performed by additional BLASTs and manual inspections of the sequences themselves. We were particularly stringent when manually curating the termite putative inserts given the existence of termite gut symbionts that are closely related to *Blattabacterium* (e.g. Bacteroides/Flavobacteria).

The genomic features in which the filtered inserts are located (such as intergenic regions, exons, introns, promoters) were identified by comparing the positions of the inserts and genomic features using bedtools and custom AWK scripts. Genomic annotation files were those previous prepared using FGENESH++.

The minimum age of ancestral inserts across the phylogeny was estimated by comparing sequence similarity of each insert and inserts in other cockroach species against the sequence similarity of each insert and the genome of the host's specific *Blattabacterium* strain. For species whose genomes have been sequenced multiple times by different research groups (i.e. *Periplaneta americana*, *Blattella germanica*, and *Zootermopsis nevadensis*), only the genome version with the most inserts was analyzed. Further, inserts part of a duplication/transposition event, as characterised above, were excluded in this dating analysis. To identify putatively ancestral inserts, the filtered inserts were first extracted from the genomes using bedtools. Second, each insert from a given cockroach genome was BLASTed against the specific *Blattabacterium* strain that the species harbours (Table S1). Only species with a published species-specific *Blattabacterium* strain were included ($n = 11$). Third, each insert was BLASTed against all of the inserts from the other cockroach/termite taxa

(excluding itself). Fourth, the two BLAST output files were compared using a custom R script. The hit was set to 'NA' if query coverage was <70%.

If sequence identity was considerably higher for the *Blattabacterium* BLAST compared to BLASTs of inserts from other cockroach/termite genomes, then the insertion was considered to have occurred subsequent to the host's divergence from the other species in the sample. In contrast, if sequence identity was higher for one or more inserts from the termite/cockroach library, than for the host's *Blattabacterium*, then the insert was considered ancestral to those species sharing it. Lastly, if sequence identities for the two BLAST hits were somewhat similar, we were unable to classify the age of the insert with confidence. Clear separations between these three cases were evident in each of the analyses (see Fig. S2). Putatively ancestral inserts were manually curated by aligning them back to the cockroach genomes from which they derived and checking for sequence synteny between the flanking regions of the cockroach genomes (see example in Fig. 2A). The divergence date of the most distantly related species both harbouring the insert was treated as the minimum age of the insert.

We characterised the source locations of the inserts along the *Blattabacterium* genome. First, we aligned the inserts back to their respective *Blattabacterium* genomes using *bwa mem*. Only species with a published species-specific *Blattabacterium* strain were included in this analysis (Table S1). Note, not all of the inserts aligned, as some inserts were characterised based on fragments of other *Blattabacterium* genomes (not only the species-specific *Blattabacterium* strain). Second, we used a combination of *samtools* and *awk* to only retain primary alignments, and to record the insert start position along the alignment. Lastly, we plotted the cumulative start position counts along the *Blattabacterium* genome using the R packages *tidyverse* v2 (Wickham et al., 2019) and *ggplot2* v3.5.1 (Wickham, 2016), whereby a linear trend indicates that the origin of the inserts in the *Blattabacterium* genome is random.

HGT insert QC

To robustly confirm that inserts were evident in raw genomic reads as well as the final assemblies, we aligned the raw reads against the respective assembled genomes. Raw reads which aligned only with the putative insert, or a subset thereof, were assessed to have likely derived from contaminant *Blattabacterium* DNA. In contrast, raw reads spanning all or some of the insert as well as 'native' cockroach flanking sequence were considered evidence that the insert was genuine (Fig. S1). This step was undertaken on five genomes. We found that none of the identified inserts aligned with solely contaminant *Blattabacterium* reads; rather, each aligned with multiple raw reads spanning the boundary between exogenous inserted DNA and cockroach flanking sequence. In addition, in cases where *Blattabacterium* contaminant DNA was found to align to the insert (alongside genuine cockroach genomic reads), the *Blattabacterium* sequences were consistently divergent from the insert sequences (Fig. S1).

To assess the sensitivity of HGT insert counts to our choice of parameters, we re-characterised inserts using alternative, less stringent thresholds (see the 'Characterising HGT inserts' section). This led to a greater number of inserts being identified (246-5841 in the cockroach genomes), though overall patterns remained

similar (Fig. S4). Note that the inserts characterised in the termite genomes using these thresholds were not manually audited, whereas the primary, more stringent dataset underwent particularly rigorous manual curation (given the presence of termite gut symbionts closely related to *Blattabacterium*).

STAR methods references

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Supplemental Information - Tables and Figures

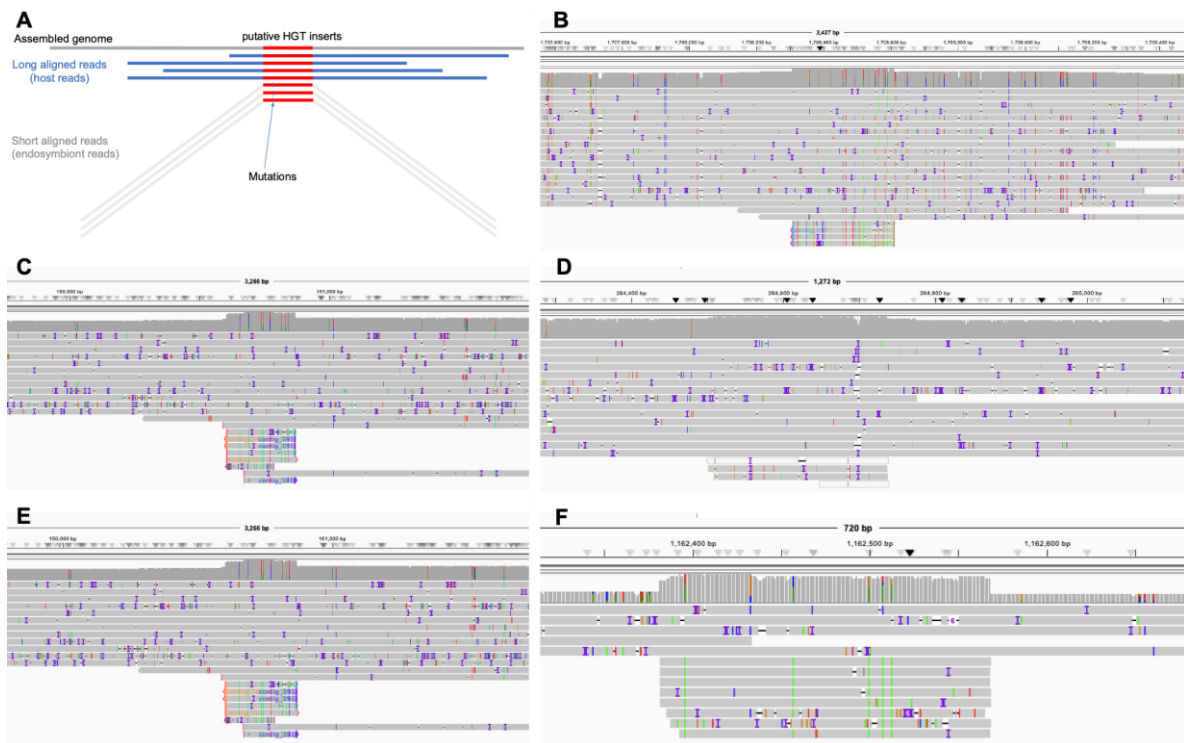


Figure S1. Schematic illustration showing methods for HGT insert confirmation (A) and examples from *M. lithgowae* (B-F). (A) The dark grey bar at the top represents the assembled genomic sequence; blue bars indicate reads broadly aligned to the genome and considered host-derived; light grey bars indicate reads aligned only at the insertion region and considered endosymbiont-derived; red marks represent the putative HGT insert sequence consistently aligned across all reads; “Mutations” denote base differences detected in short aligned reads compared with long aligned reads. Note that the “short” in “short aligned reads” refers to the length of the aligned sequence between endosymbiont and host genome, rather than the reads themselves, which were of a similar length to other long reads derived from host DNA. (B) An example of HGT insert confirmation, insert: contig_4:1708405-1708607, length of insert: 203 bp. (C) insert: contig_444:975660-975914, length of insert: 255 bp. (D) insert: contig_5553:264501-264736, length of insert: 236 bp. (E) insert: contig_8142:150463-150869, length of insert: 407 bp. (F) insert: contig_10024:1162380-1162568, length of insert: 188 bp. Due to space limitations, only raw reads aligned to the insert were displayed. The top few reads that extend beyond the image all align to both the putative HGT insert and the surrounding host cockroach sequence. The shorter ones below are assessed to have derived from contaminant *Blattabacterium* (i.e. the contemporary endosymbiont, rather than the ancestral donor of HGT inserts). Note that the *Blattabacterium* reads are longer than displayed, however large portions are not visible because they did not align to the flanking regions surrounding the insert.

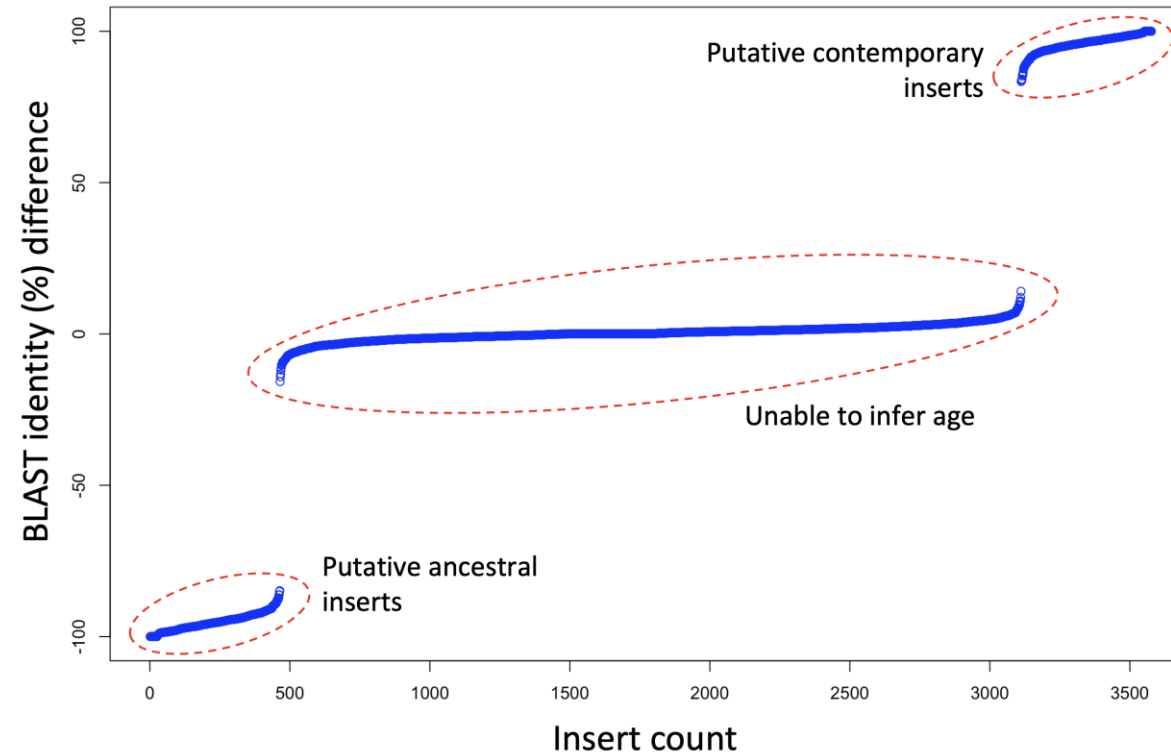


Figure S2. Example of the process to identify potentially ancestral inserts by comparing the top BLAST hit (percent sequence identity) of an insert against a *Blattabacterium* database with the top hit against a database of all other inserts (i.e. [BLAST % identity to *Blattabacterium* database] minus [BLAST % identity to the insert database]). This example shows all inserts from *Panesthia tryoni* (Eungella).

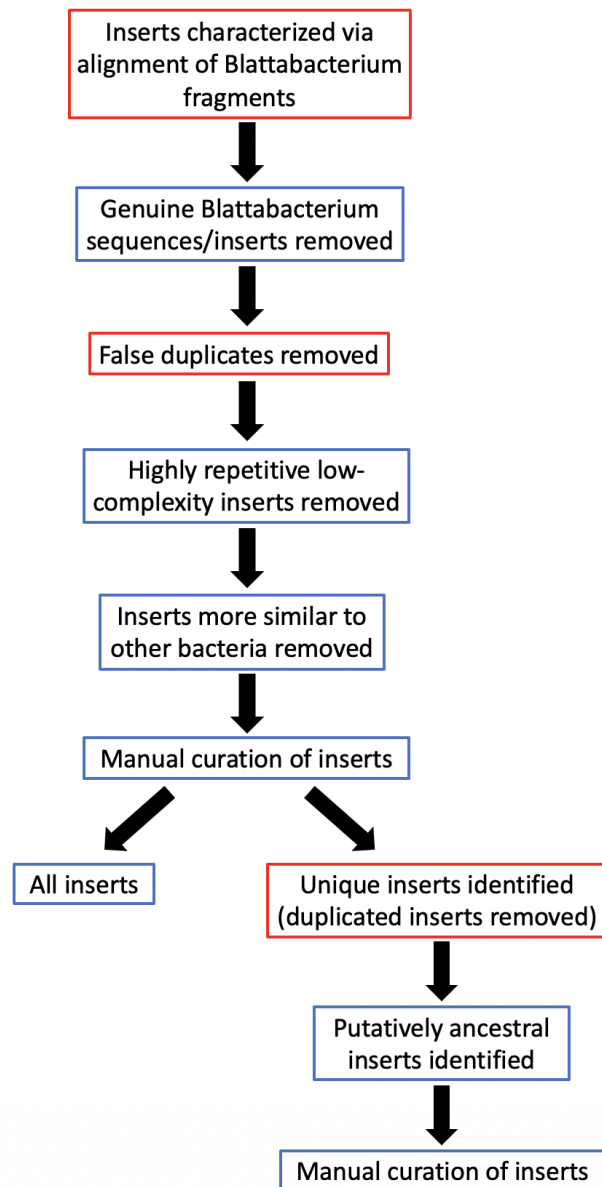


Figure S3. Insert filtering pipeline. The steps outlined with a red border indicate where alternate filtering thresholds were applied, thereby resulting in two set of inserts (with standard and alternate filtering threshold applied; the manual curation steps were not performed for alternate filtering). This pipeline workflow and the relevant scripts are available on GitHub (https://github.com/MEEP-projects/HGT_pipeline).

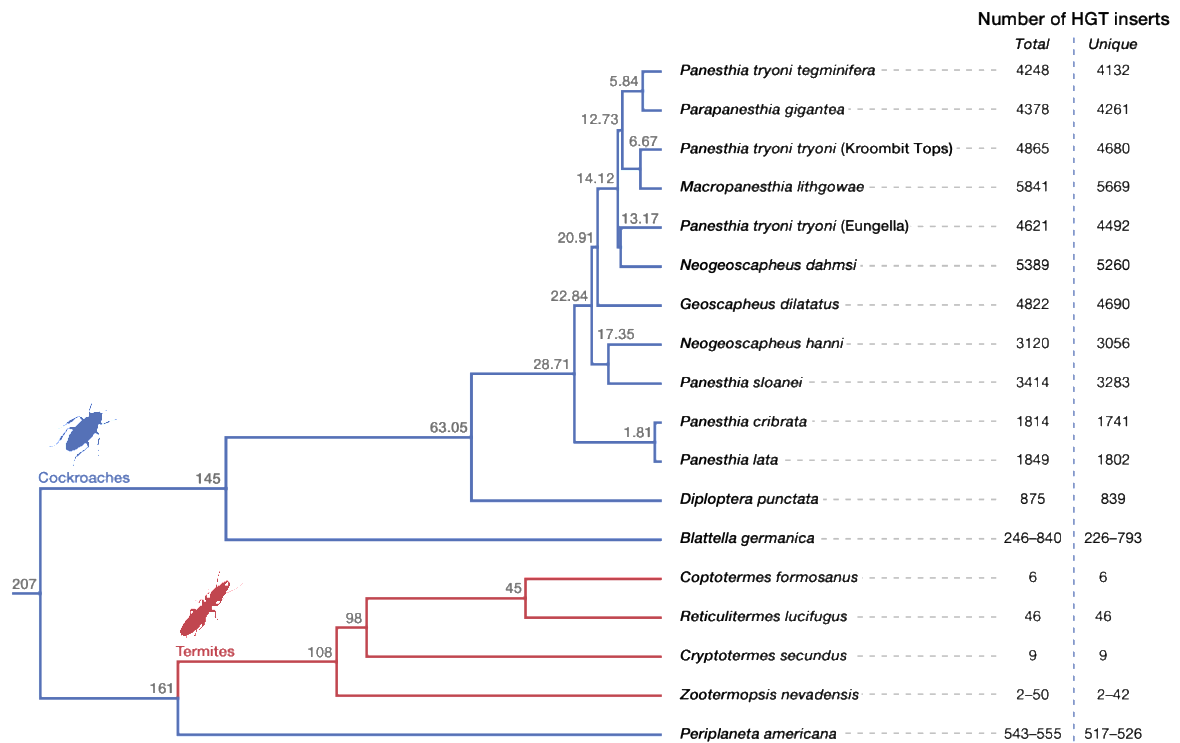


Figure S4. Chronogram of taxa analysed in this study alongside the total number of HGT inserts within each genome, and the number of unique inserts (after excluding inserts likely arising from duplication or transposition) based on the alternative, less stringent thresholds.

Table S1. Blattodea and Blattbacterium genomes details.

Species	Genome ID	Source	Species-specific Blattabacterium genome
<i>Panesthia lata</i>	M1	This study	Unavailable
<i>Panesthia sloanei</i>	Z1	This study	Unavailable
<i>Macropanesthia lithgowae</i>	Z4	This study	GCA_038291465.1
<i>Panesthia tryoni tryoni</i>	Z5	This study	GCA_038282645.1
<i>Parapanesthia gigantea</i>	Z010	This study	GCA_038286975.1
<i>Panesthia tryoni tegminifera</i>	Z011	This study	GCA_038282645.1
<i>Panesthia tryoni tryoni (Eungella)</i>	Z012	This study	GCA_038280235.1
<i>Neogeoscapheus dahmsi</i>	Z014	This study	GCA_038260915.1
<i>Panesthia cribrata</i>	T507	Ewart et al. (2024)	Unavailable
<i>Neogeoscapheus hanni</i>	4Nhan	Ewart et al. (2024)	GCA_038295885.1
<i>Geoscapheus dilatatus</i>	8Gdila	Ewart et al. (2024)	GCA_038272405.1
<i>Diploptera punctata</i>	GenBank accession: GCA_030220185.1	GenBank	CP049785.1
<i>Periplaneta americana v1</i>	GenBank accession: GCA_025594305.2	GenBank	CP001429.2
<i>Periplaneta americana v2</i>	IBG_00622	InsectBase 2.0	CP001429.2
<i>Periplaneta americana v3</i>	CNA0019281	China National GeneBank Sequence Archive (CNSA)	CP001429.2
<i>Periplaneta americana v4</i>	GenBank accession: PRJNA1098420	Re-assembled PacBio data from GenBank	CP001429.2
<i>Blattella germanica v1</i>	IBG_00115	InsectBase 2.0	CP001487.1
<i>Blattella germanica v2</i>	GenBank accession: GCA_000762945.2	GenBank	CP001487.1
<i>Zootermopsis nevadensis v1</i>	IBG_00814	InsectBase 2.0	Unavailable
<i>Zootermopsis nevadensis v2</i>	20230531.contig	This study (supplied by co-author, K. Maekawa)	Unavailable
<i>Coptotermes formosanus</i>	IBG_00204	InsectBase 2.0	Unavailable
<i>Cryptotermes secundus</i>	IBG_00218	InsectBase 2.0	Unavailable
<i>Reticulitermes lucifugus</i>	IBG_01145	InsectBase 2.0	Unavailable
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251515.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252235.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251675.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251735.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251355.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251635.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251575.1	GenBank	NA

<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251255.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252015.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252255.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251995.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251795.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251815.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252055.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251275.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252415.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251775.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251315.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252435.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252275.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252515.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251335.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251235.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251375.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252455.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251615.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251295.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251715.1	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252395.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014251695.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: GCA_014252335.1	GenBank	NA
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<i>Blattabacterium cuenoti</i>	GenBank accession: CP001429.2	GenBank	NA
<i>Blattabacterium cuenoti</i>	GenBank accession: CP003015.1	GenBank	NA

Table S2. QC of newly assembled cockroach genomes.

Sample name	Species	Location	DNA extrac tissue	Basecalling mode	Length (Gb)	N50 (Mb)	Complete BUSCOs (%)	Fragmented BUSCOs (%)
M1	<i>Panesthia lata</i>	Blackburn Island, Lord Howe, NSW	leg muscle	simplex, with sup	1.77	1.14	97.6	1.7
Z1	<i>Panesthia sloanei</i>	Mt Baldy, QLD	leg muscle and miscellaneous (not fat body)	simplex, with sup	1.82	1.67	98.7	0.6
Z4	<i>Macropanesthia lithgowae</i>	Murgon, QLD	leg muscle	simplex, with sup	1.82	1.44	98	1.1
Z5	<i>Panesthia tryoni tryoni</i>	Kroombit Tops, QLD	leg muscle	simplex, with sup	1.72	0.63	95.2	1.9
Z010	<i>Parapanesthia gigantea</i>	-	leg muscle and fat body	duplex, with sup	1.85	1.04	98.2	0.7
Z011	<i>Panesthia tryoni tegminifera</i>	Washpool, NSW	leg muscle and fat body	duplex, with sup	1.84	1.19	98.9	0.4
Z012	<i>Panesthia tryoni tryoni (Eungella)</i>	Rosehill, NSW	leg muscle and fat body	duplex, with sup	1.86	0.86	97.9	1
Z014	<i>Neogeoscapheus dahmsi</i>	Moura, QLD	leg muscle and fat body	duplex, with sup	1.9	5.33	99.5	0.1

Table S3. Number of NUMTs identified in each genome assembly. *Cryptotermes secundus* and *Diploptera punctata* were not included as their mtDNA genomes were unavailable.

Species	Genome ID	Associated mtDNA genome ID	Number of NUMTs identified
<i>Panesthia lata</i>	M1	PQ231639.1	1082
<i>Panesthia sloanei</i>	Z1	MW996597.1	1017
<i>Macropanesthia lithgowae</i>	Z4	MW354066.1	963
<i>Panesthia tryoni tryoni</i>	Z5	MW996603.1	1028
<i>Parapanesthia gigantea</i>	Z010	MW354073.1	1095
<i>Panesthia tryoni tegminifera</i>	Z011	MW996599.1	1002
<i>Panesthia tryoni tryoni (Eungella)</i>	Z012	PX580548.1	1036
<i>Neogeoscapheus dahmsi</i>	Z014	MW996587.1	1060
<i>Panesthia cribrata</i>	T507	PQ231640.1	1066
<i>Neogeoscapheus hanni</i>	4Nhan	MW996588.1	877
<i>Geoscapheus dilatatus</i>	8Gdila	MW600997.1	811
<i>Periplaneta americana</i> v1	GCA_025594305.2	GU947663.1	695
<i>Periplaneta americana</i> v2	IBG_00622	GU947663.1	708
<i>Periplaneta americana</i> v3	CNA0019281	GU947663.1	673
<i>Periplaneta americana</i> v4	PRJNA1098420	GU947663.1	715
<i>Blattella germanica</i> v1	IBG_00115	EU854321.1	261
<i>Blattella germanica</i> v2	GCA_000762945.2	EU854321.1	251
<i>Zootermopsis nevadensis</i> v1	IBG_00814	OQ559113.1	27
<i>Zootermopsis nevadensis</i> v2	20230531.contig	OQ559113.1	61
<i>Coptotermes formosanus</i>	IBG_00204	NC_015800.1	122
<i>Reticulitermes lucifugus</i>	IBG_01145	MK088051.1	150

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